Corrigé — Exercice 7 — Bac principal 2023

1a) $(3-i)^2 = 9 - 6i + i^2 = 8 - 6i$.

1b) $\Delta = (5+5i)^2 - 4(-2+14i) = 8 - 6i = (3-i)^2$, d'où $z = \dfrac{(5+5i)\pm(3-i)}{2}$:

$$S = \{\,4+2i,\ 1+3i\,\}$$

2b) $z_A - z_D = 3 - i$ et $\overline{z_D} = 1 - 3i$; $(z_A - z_D)\,\overline{z_D} = (3-i)(1-3i) = -10i$.

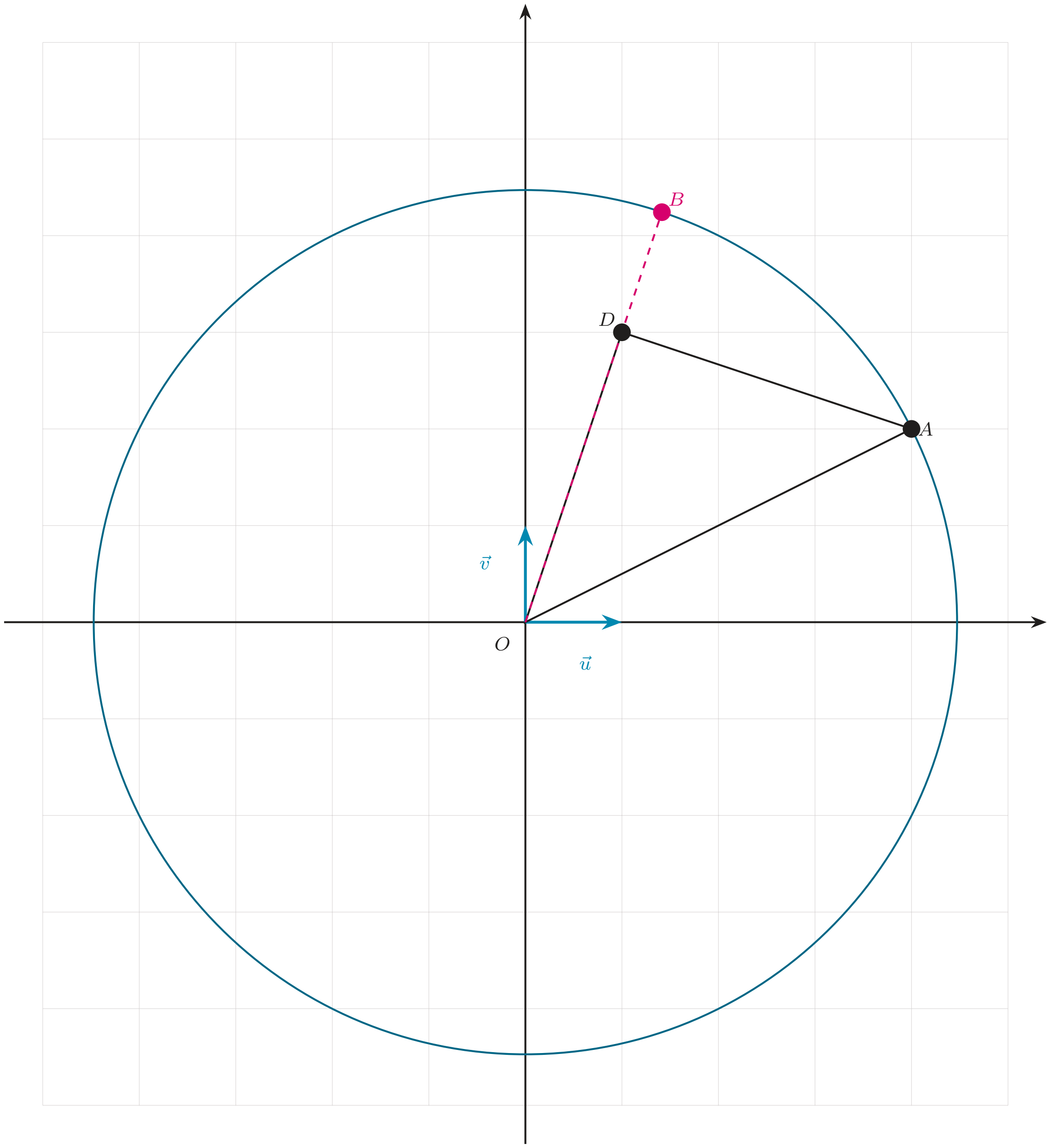
2c) $(z_A - z_D)\,\overline{(0 - z_D)} = -(z_A - z_D)\,\overline{z_D} = 10i$, imaginaire pur $\Rightarrow (DA) \perp (DO)$; et $DA = |3-i| = \sqrt{10} = |z_D| = DO$. Donc OAD isocèle rectangle en D.

3a) $|z_A| = |4+2i| = 2\sqrt{5} \Rightarrow A \in (\mathcal{C})$.

4a) $B \in (OD) \Rightarrow O, D, B$ alignés $\Rightarrow \dfrac{z_B}{z_D} \in \mathbb{R} \Rightarrow \alpha = z_B\overline{z_D} = \dfrac{z_B}{z_D}\,|z_D|^2 \in \mathbb{R}$.

4b) $|\alpha| = |z_B||z_D| = 2\sqrt{5}\cdot\sqrt{10} = 10\sqrt{2}$.

4c) $z_B = 2\sqrt{5}\cdot\dfrac{1+3i}{\sqrt{10}} = \sqrt{2}(1+3i) = \sqrt{2} + 3\sqrt{2}\,i$ (partie réelle > 0).



Corrigé Ex 7 : cercle $(\mathcal{C})$ de rayon $2\sqrt{5}$; OAD isocèle rectangle en D ; $B \in (OD)\cap(\mathcal{C})$, $z_B = \sqrt{2} + 3\sqrt{2}\,i$.